Toward Equitable AI Detection: A Hybrid Framework Integrating Stylometric Personalization and Language Proficiency Analysis

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With the surge of large language models, the education system faces challenges regarding academic integrity and student critical thinking. Current AI detectors suffer from low reliability and high bias, creating an unsustainable environment for academic evaluation. This study utilizes a dual-dataset approach, incorporating 1,272 student essays from the Uppsala Student English Corpus and 1,287 computer science abstracts from arXiv, which were paraphrased using five LLMs to create a robust human and AI-labeled dataset. The developed architecture analyzes specific writing styles, language proficiency for native and non-native speakers, and patterns associated with modern humanizers. Using a CRISP-DM and ML adversarial methodological approach, the first step was to preprocess the data, engineer features, and validate across various metrics. Results indicate that our hybrid architecture achieves an AUC of ~0.999 in controlled environments, outperforming current baselines. In real-world testing, our model reached an AUC of 0.8946, compared to the next best model at 0.828. While performance varied against modern humanizers, the architecture remained robust under strict educational thresholds (FPR < 1%). These findings demonstrate that combining stylometric, linguistic, and authorship attribution metrics reinforces AI detection. By integrating these diverse analytical dimensions, this framework provides educational institutions with a more robust and equitable foundation for maintaining academic integrity, enabling them to navigate the complexities of an AI-augmented educational landscape with greater confidence.

Keywords: AI detection, Humanizer Attacks, Large Language Models, Adversarial Paraphrasing, Personalization

1. Introduction

1.1. Background and Motivation

The continued development of large language models such as Claude, ChatGPT, Gemini, etc, has both positively and negatively affected the education system in all aspects. These models have become the fastest-growing consumer applications, reaching 100 million active users within two months of launch (Guo et al. 2023), growing every day and being integrated into every aspect of our lives. It acts as both a transformative tool for increased productivity and efficiency but poses a threat to academic integrity and the future of research and education as a whole.

In higher education, teachers tend to use AI to help plan classes and create new and innovative ways to engage students and encourage them to be more interested in the respective topic (Lo et al. 2024). Furthermore, it also helps students to study more effectively by bridging the cultural, social, or academic gaps they may have, which makes learning a more interactive experience (Mimoudi 2025). However, it can foster unethical practices that can hinder their critical thinking, decision-making, and problem-solving abilities and overall learning process (Gerlich 2025).

Particularly, the use of AI for generating texts or work provided by the educational institution can be problematic since students are primarily only looking for an efficient way to finish their assignment. They are not taking into account that the process of learning for the brain encapsulates the concept of neuroplasticity, which is the idea of restructuring connections between neurons and synapses. This can only be done by creating new neural circuits through repetition, attention, and active engagement. Consequently, this is only possible if the student completes the task by themselves with limited use of large language models to fully understand the information and skills that are trying to be taught by the school or university. Cognitive psychologist Robert Bjork coined the term desirable difficulty, which states that the process of short-term learning should be significantly harder so that in the long-term the person can better retain the information and apply it in other contexts (Bjork 1994).

With the failure of current content detectors, students are able to use AI without any limitations, which allows for an easy way forward and hinders the capability of meaningful learning. Moreover, if an AI content detector dictates that a student has used it but it ends up a false positive, meaning it misclassified this occurrence as if it were true, then it can create a lot of issues for the academic institution (Cingillioglu 2023). In conclusion, it is impossible to ignore the development and use of LLMs; however, it is also important to consider the ethical use of them in the educational aspect for the future of research and studies all around the world.

1.2. Problem Statement

The main issue with AI content detectors is their failure to consider the diverse factors contributing to various forms of academic writing. On one hand, models utilize large datasets to look for statistical patterns, estimating the likelihood of AI generation; however, there are still many accuracy issues since models average between 60–80% effectiveness with a high amount of false positives and negatives (Weber-Wulff et al. 2023). These models often classify standard academia and formal English as AI-generated by considering variables such as structure and vocabulary. In reality, this only uses a universal baseline, yet context matters, as does the fact that no one person writes the same, which is hard to quantify. This is affecting the education system because teachers cannot fully rely on these models and need to use different, time-intensive methods like problem-based learning. While there is a huge emphasis on stylometry, such as average sentence length, common function words, and semantic coherence, which is not sufficient. To increase efficiency, a more diverse architecture is needed, integrating language proficiency, analyzing bias against non-native speakers, and evaluating personalization through an author's collection of work. Furthermore, models must account for emerging humanizers that paraphrase AI text to sound human. By using a more diverse architecture for an AI detection platform, including these aspects, we can better address the limitations of current tools.

1.3. Contributions

This paper will contribute to the development of an integrated architecture that analyzes various aspects of AI content detection and can help better the current models on the market. It will act as a baseline for future AI content detection, which will help, specifically the education system, adapt to this emerging technology and allow students and teachers to be more productive and effective while also keeping their authenticity and critical thinking process intact. This paper proposes an integrated hybrid architecture that fuses linguistic features (NELA), stylometric metrics (StyleDecipher), and author fingerprinting (TRACE). It combines several other indicators relevant for AI content detection and performs better than normal tools. Additionally, it will expand research into this area, especially with humanizers, and provide a starting point for other researchers with more resources to build more robust systems to secure trust in academic institutions and education as a whole.

2. Related Work

The idea of large language models being used in education has been an emerging topic in the field, where teachers and professors rush to try to teach students the new technology that is being used, which will dictate how they will live their lives in the future. On the other hand, with the rush of AI, educational institutions are trying to reduce the use of these systems in students' work and projects, since with these new models there is no limit as to what someone can accomplish; however, there is the issue of the authenticity of work. Now, with the creation of AI content detectors, this surge of 'cheating' in the education system has become more controllable; however, recently there have been developments in the accuracy of these systems due to humanizers and the increased thinking capability of the models themselves. These content detectors use various techniques such as detecting the stylometry and semantic aspects in the writing to identify whether there are significant patterns mostly used by AI. The section below will outline existing tools for AI detection and different methods that are explored for detecting whether a text is written by AI or there is a high probability it was authentic and written by a human.

2.1. Existing Tools for AI Detection

With the introduction of large language models recently, there have been commercial and open-access detectors that have been available to the public to ensure that a person's work is original and not completely made with AI. The most common ones are tools like ZeroGPT, Grammarly, Scribbr, and QuillBot, which tend to classify the involvement of AI in writing as fully human, light contribution, heavy contribution, or fully AI-generated (Hadra et al. 2026).

At the beginning of the development of LLMs such as ChatGPT or Gemini, these detectors had a relatively unreliable rate of success when detecting AI-generated work. For instance, when using either of these tools, some often show the same percentage of AI usage or non-AI usage; however, some others vary completely and have a different answer, although most are based on the same architecture. This is due to the fact that there is a high false positive rate among answers, which means that some of the texts that are classified as AI are not actually AI, but that is because the stylometry and semantic patterns analyzed are very similar and hard to distinguish between both styles of writing (Tsigaris and Teixeira da Silva 2026; Cingillioglu 2023).

The architecture consists mainly of first using a linguistic analysis, so analyzing perplexity, the type of word the person utilizes, and burstiness of the sentence structure, since AI tends to have a similar steady rhythm in its style of writing (Hans et al. 2024). Secondly, the content detector runs this work through its own internal language model to assess whether each word and sentence would have been written by AI. After this initial analysis, the model will then break the text down into clusters of words, extract certain features from it, and compare it with their dataset, which consists of a trained machine learning model on both human-written and AI-generated content. The model will then classify the data and label the output with a probability score on how much of it was from AI or from the human itself. This method is repeatedly used in these commercial content detectors; however, some have more sophisticated methods than others. Nonetheless, they still face a large false positive rate and low accuracy, especially with the development of the humanizers, which have been able to take AI-generated texts and make them sound like a human wrote them, bypassing the algorithm that these detectors are based on (Sadasivan et al. 2023).

2.2. Stylometry

Stylometry is essentially a unique way of writing, so the use of persistent linguistic choices, such as lexical, syntactic, and structural, that establish a work of writing by a specific author. It is the statistical study of literary style and is based on the idea that people have certain habits that tend to remain consistent across everything they write and publish (Stewart et al. 2026). For instance, the number of times someone uses function words like 'and' or 'but' or the sentence length and structure that is used, so whether the person writes long and flowing sentences or short ones. As previously mentioned, this is a common method that modern content detectors are using to distinguish between AI and human-generated text.

Literature analysis shows that there are certain frameworks that are used specifically for computational stylistic analysis in the realm of AI content detectors. Namely, SALA (Stylometry Assisted LLM Analysis) is a particular modern approach that integrates quantitative features such as Type-Token Ratio (TTR), Hapax Legomenon Ratio, and average sentence length as key features in the classification process. This method uses a structured pipeline different from the black-box model of other methods and extracts the information with certain attributes and inputs it into an LLM, which acts as the judge to explain whether the data is better characterized as human or AI text (Zhang and Zhang 2026). Apart from some reviewed modern topics, there are some contributing practical frameworks.

One of them is StyleDecipher, which looks at specifically stylistic divergences by modeling discrete indicators and continuous semantic embeddings (Li et al. 2025). This means that it looks at both the computational value of the writing, for example, the number of times a comma is used and average sentence length, but also the tone, the words' semantics, and the way the ideas flow from one to another. On top of this, the NELA toolkit is also a strong tool that extracts 87 content based attributes inducing stylistic, complexity, and psychological categories, which creates an in-depth analysis of the writing (Horne et al. 2018). Lastly, there is also another theoretical method mentioned in recent studies, which is TRACE, and it is a transformer-based **contrastive embedding** that creates an author fingerprint and remains stable under paraphrasing (Wang et al. 2024). The method identifies the style of writing through semantic tracing. Evidence-based reporting (specific phrases or structural choices) and cross-model identification, which identifies between AI models and human writing. This is then transformed into a fingerprint for a specific author, which can be used to check other new writings and see if they match their own personal style. The only issue with this is that it is really complex and requires a lot of computational power, so it is just mainly theoretical and not used in a real-world context. Overall, there has been a lot of development for models to analyze stylometry within AI-generated text, with some being extremely precise and can account for personalization of the specific author; however, they are not combined with any other method for AI detection.

2.3. Language Proficiency Analysis

The introduction of large language models has especially helped non-native speakers of many languages, but specifically English. The use of AI has helped them to overcome some challenges, for example, vocabulary usage, sentence formation, and many other things. However, with the development of these content detectors, many of these people are facing issues regarding a high probability of AI generation in their writing due to bias and misclassification. For instance, essays written by non native speakers are classified as AI-generated 61.3% of the time, while for native speakers this only accounts for 5.1% of the time (Liang et al. 2023).

Research shows that this bias comes from the content detector's reliance on perplexity as a measure of word predictability (Liang et al. 2023). Since non native speakers have lower linguistic variability, meaning that they have limited word choices and tend to use the same words repeatedly across the text, this is similar to the low perplexity patterns that are produced by most LLMs. As a result, non native speakers are incorrectly flagged with a large amount of AI usage, which makes it hard for them to adapt because they are limited by the vocabulary and their respective level of English proficiency.

On the other hand, there is also genre sensitivity for content detectors, where accuracy is also context-dependent, so for instance humanities or science-related writing differs due to the use of technical language and high lexical density that closely resembles AI outputs (Jiang et al. 2024). This can also increase the detection probability of AI usage since more vocabulary, especially technical and complex, with simple or standard sentence structure can mirror text generated by a large language model. Consequently, this is also causing authors to make their original work more chaotic with an increase in filler words or changing sentence structure due to false AI detection from these models.

2.4. Humanizers and Paraphrasing

An increase in advanced tools such as QuillBot, Clever AI Humanizer, and other paraphrasers that change AI-generated text and turn it into human-style writing are being used to combat the originality checks by the AI detectors (Muneer et al. 2025). The success rate for these humanizers often reaches over 85% (Cheng et al. 2025) and challenges the narrative of AI usage because students, especially, are able to get away with doing their work solely by using AI without getting caught. The humanizers work by swapping certain words using synonyms and use sentence restructuring in order to drop the mathematical probability for it to appear as AI-generated text. This is done by using homoglyph attacks, which involve changing certain characters into other similar ones; for instance, the letter 'e' from English to the one in Cyrillic so that the detector fails to classify this as a registered letter, and it breaks the algorithm so the percentage of AI usage is way lower (VIGILANT 2023).

More importantly, the humanizers mostly utilize backtranslation, which is changing the text through multiple languages and back to English to create certain shifts that move the flow of the text and distort the AI's signature pattern (Muneer et al. 2025). This can also work by changing sentence structure and the use of different words in order to sound more natural and human-like. This is known as a guided recomposition, whereby using an LLMs reasoning it can suggest different methods such as stylistic changes (sentence length) or use of function words (Shokri et al. 2025). All of this work by the humanizers has become really hard to track by current AI detection models and therefore, there are a lot of false positive rates and the accuracy has fallen significantly. It is vital to find a way to target the work of the humanizers and reverse engineer it in order to reduce the amount of AI usage especially for the educational field.

2.5. Synthesis and Gap Identification

In order to investigate the newly proposed features in a redesigned architecture for AI content detectors it is important to first define the main research question for this paper as the following: "How can a multidimensional AI content detection framework integrating stylometric personalization, language proficiency, and a semantic encoder reduce the low accuracy and false positive rate of current AI detection tools?" As stated in the related work, modern detectors only focus on general statistical measures for AI content detection; however, there are many other tools that incorporate other aspects. By using these tools and fusing them into one framework, it can theoretically help to increase the efficiency of AI content detection, which is going to be most beneficial for academic institutions. The rest of this paper reviews related work, builds on the methodology used for preparing the data, building the architecture accordingly, discussing the results and any key findings made. Additionally, it highlights practical applications for the study and limitations that were present.

3. Methodology

This study will use a hybrid methodological approach incorporating both the CRISP-DM framework (Wirth and Hipp 2000) and an adversarial ML evaluation development for significant validation and robustness assessment.

3.1. Problem Formulation

To understand the problem and the aim of the research clearly, the focus of this study centered around two main sub-questions. The main issue is that current AI content detectors are inefficient in determining whether a piece of text is generated using AI or not. By dissecting this issue, it was evident there were two main aspects to it, which were based on the proposed architecture and humanizer attacks. Firstly, how does the newly developed combination of feature-based extractors such as NELA, StyleDecipher, and TRACE perform compared with modern detectors under out-of-distribution academic abstracts. Secondly, the focus is on how humanizers affect overall precision and accuracy in modern detectors. Using both of these specific contexts, the study examines how the proposed architecture performed under the strict $FPR \leq 1\%$ threshold. This is due to the fact that in academia there is a pattern where texts are misclassified as AI during submission.

3.2. Dataset Construction

The study consists of using two existing datasets; one contains 1,272 human-written essays from students at Uppsala University in Sweden (Axelsson 2000) and the other is an arXiv set with 1,287 abstracts from certified journals in the Computer Science realm. One of these datasets were then rewritten using a variety of LLMs including Gemini 2.0 Flash Lite (Google), Mistral Small (Mistral AI), LLaMA 3.1 8B Instant (Meta via Groq), Command R+ (Cohere), and Claude Haiku 4.5 (Anthropic), with average 4.76 rewrites per texts due to some restrictions of some LLMs, and for second only Claude Haiku 4.5 (Anthropic) is used, in order to create both a human and AI label for further testing. Furthermore, the requirement for the dataset was also to include two or more human texts per author in order for the TRACE-based extractor to function. There were also some API limits and quotas that had to be met during the entire process; hence, there are fewer texts in one dataset than in the other. Consequently, the first dataset provided was shortened down to 963 human and 4,584 AI-paraphrased texts. As shown with the numbers in both datasets, there is one limitation where one was able to be rewritten by multiple LLMs while the second one managed to be only rewritten by one, which may have hindered the results a bit in the end.

3.3. Proposed Architecture

For the purpose of improving current AI detectors using a combination of different features, the study will use a newly proposed architecture backed up by the research done in the related work section. It was identified that the current detectors were only focusing on one aspect, for instance, stylometry, and did not consider, for instance, personalization and language proficiency analysis. In order to improve the efficiency and accuracy of the detectors, the new version integrates three extractors (NELA, StyleDecipher, TRACE) based on linguistic features, style, and author fingerprinting. By using all of these together, they are combined in 11 different methods; 4 of them have different fusion methods with a feed-forward neural network classifier, and 7 of them include different classical ML classifiers previously concatenated, as seen in the diagram below. Each of the 4 fusion models uses a two-layer feed-forward classification head with a ReLU activation function and a dropout of 0.3. This works by using different fusion techniques such as concatenation, MLP, self-attention, and scalar gating

(Neverova et al. 2016). On the other hand, the other 7 classical classifiers are standard ML such as XGBoost, Logistic regression, and Random forest methods that are concatenated to 225-dimensional flat vectors. In the end, the output generated consists of two labels of 0/1 indicating whether the text is human or AI. Moreover, it provides the confidence score of each label accordingly.

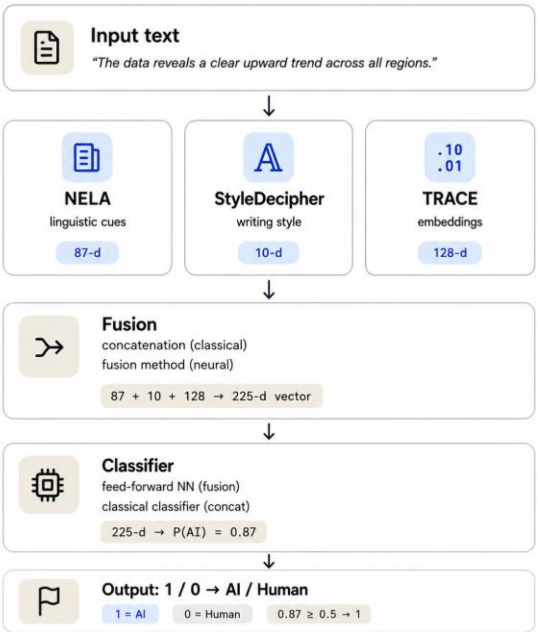

Figure 1: Proposed Architecture

3.4. Feature Extraction and Model Development

To support the task of feature extraction and model development, NELA, StyleDecipher, and TRACE were used in the process. The dimensionality of each feature extractor is explained subsequently. NELA has 87 attributes such as TTR (type-token ratio), word count, and stop-word rate, which show various statistics regarding linguistic features that can differ between AI and human-written texts (Horne et al. 2018). The StyleDecipher-based extractor rewrites the texts using certain LLM models, creating five metrics to differentiate between AI and human texts, therefore, used later in the testing phase. These metrics capture shared vocabulary, local word ordering, short phrase preservation, character overlap, and semantic similarity (Li et al. 2025). The last extractor is inspired by a TRACE contrastive learning framework which was adapted into a tactical method for this architecture (Wang et al. 2024). The input consists of evaluated text and a specific number of other authors' texts (>0), which is then segmented into individual sentences and scored through the TF-IDF formula. Subsequently, it keeps only 5 highest scored sentences per text, which, afterwards, passes through the AnnaWegmann style embedding (a RoBERTa-based model that is contrastively fine-tuned using Reddit texts), outputting a 768-dimensional space (Wegmann et al. 2022). Same author - closer embedding space, different author - further in embedding space, which greatly measures style distance. After that, the model uses fixed random projection ("Johnson-Lindenstrauss projection") to convert 768 dimension embedding space to 128 dimensions for dimensionality reduction and preserving distances. After all, by means of pooling each sentence, the final output is a single 128-dimensional embedding space. It means the TRACE-based method shows the average author fingerprint in the 128-dimensional space. The model is then built on all of these feature extractors as previously mentioned in the architecture.

3.5 Evaluation and Validation Protocol

For effective evaluation of the models and the entire architecture, various metrics such as AUC, accuracy, F1 score, and a threshold of less than 1% for FPR rate. The major indicator employed in all of the models was AUC-ROC (Area Under the Curve), which evaluates how well a detector ranks AI-written text over human text across all thresholds. Other indicators such as accuracy and F1 score were used to ensure that the models were appropriately measured and their performance was adequate for this type of research. In order to evaluate the detectors, we split the dataset in two different ways using these metrics. The dataset was divided into an out-of-distribution and controlled split to ensure that the data was trained and validated all throughout the process.

The controlled split was chosen to ensure that our models work for a standard, strict set in order to proceed to the out-of -distribution experiment, which simulates real-world testing. For the controlled set of splits, the USE dataset was used with a 70/15/15 proportion randomly assigned. This means that 70% of the data was the training set, and the other corresponding 15% was used for validation and testing. The results from this split showed that there was state-of-the-art performance from the proposed architecture. Correspondingly, the out-of-distribution split used a 90/10 proportion, where 90% was used for training and the other 10% for validation; to test this, the arXiv dataset was used. Moreover, humanizers were used to rewrite the AI-generated texts, and this was also tested using the same detectors for robustness. Altogether, the chosen splits allowed the research to be effectively analyzed and usable for further studies.

4. Results

4.1. Baseline Performance for Controlled Environment

In accordance with the controlled split of the dataset, a total of 14 detectors were used for concatenation, including 4 fusion models, 7 classical classifiers, and 3 separate models (NELA/StyleDecipher/TRACE). The separate models used only Random Forest Classifiers with a number of estimators of 400. All of the combination models performed with state-of-the-art results (macro f1: 0.9853-0.9979; AUC: ~0.999), with the best model being logistic regression.

The NELA extractor showed relatively good results (macro f1: 0.9908; AUC: 0.9998) for this controlled set. All the TPR at FPR<=1 rates were successfully good: around 0.97-0.99 for each method. TRACE and StyleDecipher separate classifiers showed relatively worse results (macro f1: 0.7914 and 0.6230; AUC: 0.9374 and 0.8778; TPR at FPR<=1%: 0.5326 and 0.4208 accordingly). This suggests that the NELA feature extractor was the most efficient and strongest and outperforms many of the current detectors on the market, whilst TRACE and StyleDecipher are weaker individually. Merging these three different extractors creates a synergistic effect that, as a result, helps achieve a lower false positive rate and more accurate outcomes.

4.2. Multimodal Models vs. Published Baselines

With the out-of-distribution split, the most proficient classifier out of all of them turned out to be logistic regression for the combinational architecture. This was due to a massive overfitting issue with the other respective classifiers, which made the simpler one more effective in the end. The following graph displays the AUC values from each of the detectors that were analyzed in our research. The proposed architecture performed here with an AUC of 0.8946, with the next best model being DetectGPT (N=10, AUC=0.828) and the gated fusion method.

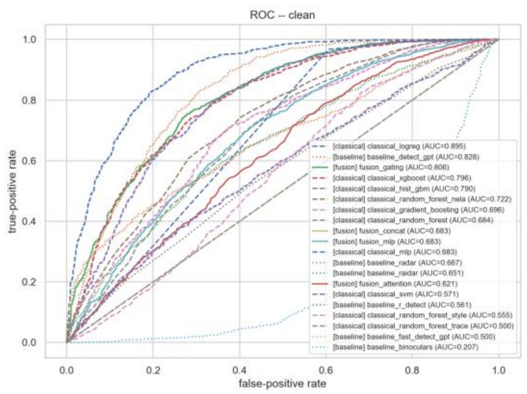

Figure 2. ROC Curve for the Out-of-Distribution Test Split

(AUC=0.8063). The slope tends to rise sharply early on, which indicates that the model has a low false positive rate early on. This shows that the model or combinational architecture is able to have a relatively high true positive rate before flattening out when it reaches higher error thresholds. In terms of the controlled testing, the scores from the different models were nearly perfect in comparison to applying it to real-world data. This means that the architecture worked well in a testing environment and was validated accordingly so that it could outperform other models when tested with the dual dataset collected, which included essays and abstracts.

4.3. Robustness under Adversarial Humanization

In addition to the analysis performed with the newly proposed architecture and current detectors, the study was also extended to look at humanizers. The AI-generated texts were paraphrased using two distinct humanizers (Adversarial and Temp) accordingly and then analyzed through the splits (Cheng et al. 2025; Huang et al. 2025). The results are displayed below on the ROC curve. The use of humanizers decreased the efficiency of our model to distinguish true positive examples, with an AUC of 0.819. Consequently, all

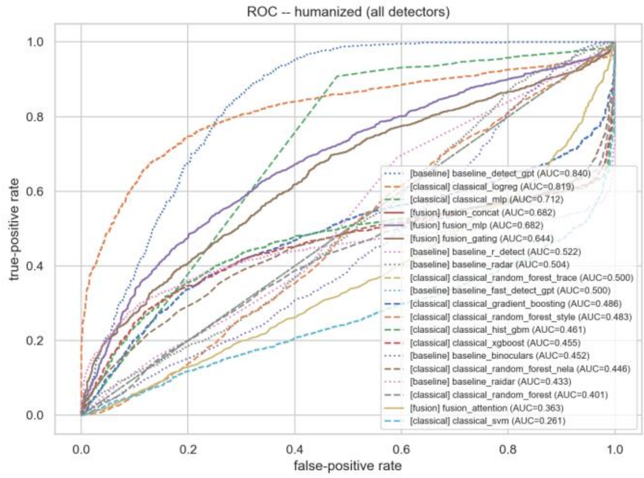

Figure 3. ROC Curve for Humanized Texts

the other models also performed worse with the addition of the humanized tests, as seen on the graph, due to their ability to mask AI-generated text as human written one using complex methods. Interestingly, the DetectGPT performed better than the proposed architecture with the humanizers, resulting in an AUC of 0.840. This means that this detector was engineered to be more robust and vigilant for certain characteristics in relation to modern-day humanizers, even if its efficiency for standard detection was not as high as the main model. For instance, this commercial model can be optimized for specific backtranslation or synonym swapping techniques rather than stylistic authenticity when it comes to AI content detection. Overall, this analysis proves that humanizers are no longer just simple tools to rewrite AI-generated text but are more complex and actively try to confuse standard semantic analysis. The drop in performance shows that even by using multi-feature frameworks such as the one developed in this paper, the foundational style markers such as lexical density can be smoothed out by paraphrasing tools.

5. Discussion

5.1. Interpretation of Key Findings

The combinational model generally outperformed other detectors, leading our team to conclude that combining techniques is effective and warrants further investigation. Training and evaluation on the controlled dataset yielded state-of-the-art performance, which supports this study's core assumption. As the NELA extractor outperformed all other techniques, we suggest that further development of statistical methods for text analysis is necessary, as these remain robust against LLM paraphrasing. The TRACE-based method performed well only within combination models, indicating that more research regarding authorization techniques should be applied. The challenges with the TRACE-based method stemmed from both limited data and embeddings that need to be trained on educational materials rather than the Reddit texts used in our current Wegmann-style embeddings (Wegmann et al. 2022). Addressing these data limitations and refining the embedding training will improve the overall accuracy of the model. A key finding is that the out-of-distribution (OOD) dataset presents a significantly different set of results compared to the controlled dataset. One primary factor to consider is the impact of humanizers on detection results; therefore, future studies should rely more heavily on widely used humanizers, as they are crucial for testing the robustness of new methods.

5.2. Implications for Current Detectors

Current detectors performed poorly on the out-of-distribution (OOD) dataset, which supports our initial premise. Certain techniques, such as the StyleDecipher-based extractor, yielded lower results than anticipated when compared to the NELA-based extractor. Notably, all models failed when evaluated under the metric requiring a false positive rate (FPR) of 1% or less, highlighting the inadequate preparation of currently available detectors. The only model that demonstrated resilience, and even improved results, during the paraphrasing process was DetectGPT, a perturbation-based method (Mitchell et al. 2023). Conversely, other models succumbed to humanizing vulnerabilities, including the combination model that integrates adversarial and statistical methods, the discriminative RADAR (Hu et al. 2023), and the kernel-based R-DETECT technologies (S. Zhang et al. 2024).

5.3. Limitations

This study, however, did have several limitations that hindered its ability to produce more significant results in relation to classifying AI-generated content. A significant limitation of this study is the scarcity of multi-author datasets within the educational domain, which constrains the generalizability of our results to real-world applications. The dataset also lacks diversity in other academic disciplines and is not large enough in comparison to the other commercial models, which could also indicate another reason why DetectGPT performed better with the humanized sample. Beyond data availability, we faced substantial resource and time constraints. Specifically, the process of generating AI samples and extracting StyleDecipher features using five different 7–8B open-source models, alongside testing various proposed configurations for the TRACE-based extractor, which required several weeks of compute time. The score of our humanizers was also limited to only using two types instead of many different ones to gauge how different paraphrasing tools work as a whole. Consequently, these constraints limited our ability to conduct more exhaustive research into the proposed methods or to utilize more advanced LLMs for evaluation across more complex datasets.

5.4. Methodological Contributions

This study proposes and seeks to advance a unified, three-modal architecture that fuses statistical-linguistic (NELA), stylometry-based (StyleDecipher), and author-fingerprint (TRACE) extractors into a single representation. While this framework is not yet ready for production or deployment, it serves as a robust foundation; its most effective configuration despite being trained on a limited dataset and achieved a higher AUC on the out-of-distribution (OOD) test set than the other detectors evaluated. Our research confirms that the theoretical, contrastive TRACE-based method contributes significantly to this architecture. Consequently, we propose specific practical improvements for its future implementation. Currently, the Wegmann RoBERTa embeddings (Wegmann et al. 2022) are trained on Reddit messages rather than formal academic texts; therefore, retraining embeddings on academic corpora is essential for better performance. Furthermore, we propose replacing the Johnson-Lindenstrauss projection (which reduces dimensionality from 768 to 128) with a projection learned jointly via contrastive examples. This shift will ensure that dimensionality reduction preserves stylistic directions rather than merely maintaining raw distances. A primary contribution of this study is the benchmarking of detector robustness against two popular modern humanizing techniques. These tests revealed that, with the exception of perturbation-based families like DetectGPT, most detectors fail under paraphrase attacks. We also established a strict, educationally motivated evaluation protocol requiring a false positive rate (FPR) of less than 1%. Our findings demonstrate that all current state-of-the-art detectors fail to meet this threshold, leading to the conclusion that none are currently prepared for the detection of academic texts in real-world, out-of-distribution environments. Finally, we have released two generated datasets: USE essays and arXiv abstracts, which were labeled for AI detection to facilitate future research into personalization-based methods that require multiple samples per author.

6. Conclusion

In conclusion, this study examined three modern approaches to LLM-generated text detection (NELA, StyleDecipher, and TRACE-based extractors) both in isolation and as a fused architecture, to assess whether combining these techniques improves detection capabilities. We found that the NELA linguistic statistical extractor was the most dominant performer compared to both the style-based StyleDecipher and the theoretical TRACE-based authorship module. Although the TRACE-based method was previously viewed as a purely theoretical approach, our implementation showed that it provides significant value when integrated into a fused model, confirming that author personalization is essential for the future of detector design.

The core contribution of this work lies in addressing critical performance limitations and the impact of humanizers. Under our strict educational threshold for identifying non-human-authored texts, every modern detector and extractor failed when tested on real-world, out-of-distribution samples. Furthermore, humanizers effectively dismantled the accuracy of nearly all tested detectors, with only the perturbation-based model demonstrating any resilience. These findings are critical because educational institutions currently rely on detectors whose performance in controlled settings does not translate to real-world applications, placing students at risk of being falsely accused of AI-assisted cheating. By defining a strict false-positive evaluation protocol and providing a reproducible, three-extractor architecture that outperforms existing tools on out-of-distribution splits, this paper offers researchers and institutions a clearer understanding of the current limitations in detection tools and establishes a foundation for future architectural improvements.

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